$\boxed{T6}$ $p(x) = \begin{cases} \frac{\theta-1}{x^{\theta}}, & x \geq 1 \\ 0, & x < 1 \end{cases}$ $\theta > 1$

a)

<u>ОНП:</u>

$L(\theta) = \frac{(\theta-1)^n}{\prod_{i=1}^{n} x_i^{\theta}} = \prod_{i=1}^{n} p(x_i, \theta)$; $\ln L(\theta) = n \ln(\theta-1) - \theta \sum_{i=1}^{n} \ln x_i \Rightarrow$

$\Rightarrow \frac{\partial \ln L(\theta)}{\partial \theta} = \frac{n}{\theta-1} - \sum_{i=1}^{n} \ln x_i = 0$

$\Rightarrow \frac{n}{\theta-1} = \sum_{i=1}^{n} \ln x_i$; $\frac{1}{n} \sum_{i=1}^{n} \ln x_i = \frac{1}{\theta-1} \Rightarrow \underbrace{\tilde{\theta} = 1 + \frac{1}{\frac{1}{n}\sum_{i=1}^{n} \ln x_i}}$

$\frac{\partial^2 \ln L(\theta)}{\partial \theta^2} = -\frac{n}{(\theta-1)^2} < 0 \quad \forall \theta > 1 \Rightarrow$ лок. максимум.

b) ① $\Theta = (1; +\infty)$ - открытое

② Сильно-регулярная модель; <u>2.1. Регулярность</u>:

- $p(x, \theta)$ - непр. дифф по θ на $(1; +\infty)$
- $\int_1^{+\infty} \frac{\partial p(x,\theta)}{\partial(\theta)} dx = \int_1^{+\infty} \frac{\ln x - \theta \ln x + 1}{x^{\theta}} dx =$

$= \frac{x^{1-\theta}}{1-\theta}\Big|_1^{+\infty} + \int_1^{+\infty} \ln x \, dx^{1-\theta} = \frac{1}{\theta-1} - \frac{1}{\theta-1} = 0$

- $I(\theta) = M\left[\left(\frac{\partial \ln p(x,\theta)}{\partial \theta}\right)^2\right] = \int_1^{+\infty} \frac{\theta-1}{x^{\theta}} \left(\frac{1}{\theta-1} - \ln x\right)^2 dx = \int_1^{+\infty} \left(\frac{1}{x^{\theta}(\theta-1)} - \right.$

$\left. - 2\frac{(\theta-1)\ln x}{x^{(\theta)}(\theta-1)} + \frac{(\theta-1)\ln^2 x}{x^{\theta}}\right) dx = \frac{1}{(\theta-1)^2} - 2\int_1^{+\infty} \frac{\ln x}{x^{\theta}} dx + \int_1^{+\infty} \frac{(\theta-1)\ln^2 x}{x^{\theta}} dx$

$-2\int_1^{+\infty} \frac{\ln x}{x^{\theta}} dx = 2\frac{1}{(1-\theta)^2}$; $\int_1^{+\infty} \frac{(\theta-1)\ln^2 x}{x^{\theta}} dx = -2\frac{1}{(1-\theta)^2}$

$I(\theta) = \frac{1}{(\theta-1)^2} + \frac{2}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} = \frac{1}{(\theta-1)^2}$ - непр на $(1; +\infty)$ и > 0

$\Rightarrow$ модель <u>регулярна</u>

<u>2.2</u> $\frac{\partial^2 p(x,\theta)}{\partial \theta^2} = \frac{-2\ln x + \ln^2 x(\theta-1)}{x^{\theta}}$; $\frac{\partial^2}{\partial \theta^2} \int_1^{+\infty} p(x,\theta) dx = 0$

$\int_1^{+\infty} \left(-\frac{2\ln x}{x^{\theta}} + \frac{\ln^2 x(\theta-1)}{x^{\theta}}\right) dx =$

$= \frac{2}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} = 0$

<u>2.3</u> $p(x,\theta)$ - 2-жды непрер. дифф. на $\Theta = (1; +\infty)$

$\Rightarrow$ Модель сильно регулярна

$\Rightarrow$ по (1) ОМП - состоятельная, асимпт. несмещенная, асимпт. нормальная асимпт. эффективна.

$p = 0{,}5$ (медиана) ; $\int_1^x \frac{\theta-1}{t^{(\theta)}}dt = \frac{-1}{x^{\theta-1}} + 1 = \frac{1}{2}$

$F(x_p) = p$; $F(x_{0,5}) = 0{,}5$

$x_{0,5} = 2^{\frac{1}{\theta-1}} \Rightarrow g(\hat\theta) = 2^{\frac{1}{\hat\theta-1}}$; $g(\theta) = 2^{\frac{1}{\theta-1}}$

$\nabla g(\hat\theta) = \cancel{2^{\frac{1}{\hat\theta-1}}\ln 2} = 2^{\frac{1}{\hat\theta-1}} \cdot \ln 2 \cdot \left(-\frac{1}{(\hat\theta-1)^2}\right)$

$I^{-1}(\theta) = (\theta-1)^2 \Rightarrow \sigma^2(\theta) = \nabla^T g(\theta) \cdot I^{-1}(\theta) \cdot \nabla g(\theta)$

$\sigma^2(\theta) = \left(2 \cdot \frac{1}{\theta-1} \cdot \ln 2 \cdot \left(-\frac{1}{(\theta-1)^2}\right)\right)^2 \cdot (\theta-1)^2$

$\Rightarrow \sqrt{n}\,\frac{\left(2^{\frac{1}{\hat\theta-1}} - 2^{\frac{1}{\theta-1}}\right)}{2^{\frac{1}{\hat\theta-1}} \cdot \ln 2 \cdot \frac{1}{(\hat\theta-1)}} \sim N(0;1)$ $\quad t_1 = U_{\frac{1-\beta}{2}}$

$t_2 = U_{\frac{1+\beta}{2}}$

$$P\left(t_1 < \frac{\sqrt{n}(\hat\theta-1)\left(2^{\frac{1}{\hat\theta-1}} - 2^{\frac{1}{\theta-1}}\right)}{2^{\frac{1}{\hat\theta-1}} \cdot \ln 2} < t_2\right) = \beta$$

$$\Rightarrow \frac{2^{\frac{1}{\hat\theta-1}} \ln 2}{\sqrt{n} \cdot (\hat\theta-1)} t_1 < 2^{\frac{1}{\hat\theta-1}} - x_{0,5} < \frac{2^{\frac{1}{\hat\theta-1}} \ln 2}{\sqrt{n}(\hat\theta-1)} t_2$$

$$\boxed{2^{\frac{1}{\hat\theta-1}} - \frac{2^{\frac{1}{\hat\theta-1}} \ln 2}{\sqrt{n}(\hat\theta-1)} t_2 < x_{0,5} < 2^{\frac{1}{\hat\theta-1}} - \frac{2^{\frac{1}{\hat\theta-1}} \ln 2}{\sqrt{n} \cdot (\hat\theta-1)} t_1}$$

c) $\sqrt{n} \cdot \frac{\hat\theta - \theta}{\sigma(\hat\theta)} \sim N(0;1)$ $\quad g(\theta) = \theta \Rightarrow \nabla g(\theta) = 1$

$\sigma^2(\theta) = 1 \cdot (\theta-1)^2 \cdot 1$

$\sqrt{n}\,\frac{\hat\theta - \theta}{(\theta-1)} \sim N(0;1)$ $\quad \hat\theta = 1 + \frac{1}{\frac{1}{n}\sum_{i=1}^n \ln x_i}$

$\sqrt{n} \cdot \frac{1}{n}\sum_{i=1}^n \ln x_i (\hat\theta - \theta) \sim N(0;1) \Rightarrow \hat\theta - 1 = \frac{1}{\frac{1}{n}\sum_{i=1}^n \ln x_i} \cdot \frac{1}{\hat\theta-1} = \frac{1}{n}\sum_{i=1}^n \ln x_i$

$t_1 = U_{\frac{1-\beta}{2}}$

$$P\left(t_1 < \frac{1}{\sqrt{n}}\sum_{i=1}^n \ln x_i (\hat\theta - \theta) < t_2\right) = \beta$$

$t_2 = U_{\frac{1+\beta}{2}}$

$$\boxed{\hat\theta - \frac{t_1}{\frac{1}{\sqrt{n}}\sum_{i=1}^n \ln x_i} < \theta < \hat\theta - \frac{t_1}{\frac{1}{\sqrt{n}}\sum_{i=1}^n \ln x_i}}$$ ~ асимпт. доверит. интервал. по ОМП.